Resilient Emergency MANETs for Wilderness Safety

Supplementary Materials: Artifact Guide, Reproduction Instructions, and Extended Results

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This supplement accompanies *Resilient Emergency MANETs for Wilderness Safety: Directed Study Final Report* (2026-07-26). Section 1 maps every artifact the report cites to its location in the project repository (`github.com/edoherty2017/resilient-emergency-manets`, branch `trial2-field-campaign-2026-07` — the canonical version of all material). Section 2 gives verified commands for running the simulator. Sections 3–4 present extended views of the released year-scale results. All simulation values are MODEL-ONLY (uncalibrated), drawn exclusively from the released `release_v1` aggregates.

1 Guide to the artifact repository

Report section	Repository location
§5.2–5.3 year-scale results	`artifacts/sim/corrected/release_v1/` (`corrected_stats.json`, `meaningful_metrics.json`; locked inputs and hashes alongside)
§4 parity, reproducibility, tests	`artifacts/sim/corrected/` (`micro_parity_2026-07-17.json`, `repro_check_2026-07-17/`, `cargo_test_2026-07-26.log`)
§5.1 Trial 1 and ITM screens	`artifacts/airmap/live_trial/quality_gates.json`; `artifacts/coverage_prediction/`; `artifacts/itm/`
§6 Trial 2 registrations	`artifacts/trial2/` (prediction packs, `prereg_manifest.json`) and the dated siting documents in `docs/`
§6 field raw evidence	`artifacts/trial2/raw_pull_20260721/`, `raw_pull_20260726/` (SHA-256 checksums within)
§6.6 station-cadence evidence	`artifacts/trial2/nhmesh_activity_20260726/`
Raw dataset release	`artifacts/dataset_release/evidence-2026-07-26/`

Table 1: Report citations to repository paths. Registrations are git-timestamped by commit `1229309`; every SHA-256 digest cited in the report re-verifies with `shasum -a 256`.

The 2026-07-17 report draft and the `WITHDRAWN-DO-NOT-CITE/` directory are retained archives only; project claims should not be reviewed from them.

2 Running the simulator

Requirements: a Rust toolchain (`rustup`) for the release engine; Python 3.12+ with the repository virtual environment for the reference engine and analysis scripts. All commands run from the repository root and were executed successfully on 2026-07-27.

Build.

```
(cd fastsim && cargo build --release)
```

Single run. One simulated day completes in roughly two seconds; a full 365-day statewide year in about 3.5 minutes:

```
./fastsim/target/release/fastsim \
  --topology artifacts/sim/topology_statewide.json \
  --routes   artifacts/sim/routes_statewide.json \
  --weather  artifacts/sim/weather_year.json \
  --config   config/sim/wmnf_sim.yaml \
  --mode duty_sync --days 365 --seed 42 --out /tmp/year.json
```

The `--mode` argument accepts: `flood`, `min_hop`, `etx`, `energy_aware`, `lb_energy`, `duty_sync`, `duty_adaptive`, `rotate_lb`, `selective_duty`.

Reproduce the released sweep. Nine modes × seeds 42–46 with hash-locked inputs and an aggregate manifest:

```
scripts/run_corrected_release.sh /tmp/my_release
```

Compare `/tmp/my_release/corrected_stats.json` against `artifacts/sim/corrected/release_v1/corrected_stats.json`.

Verification suite.

```
(cd fastsim && cargo test)                    # 47 + 3 tests
.venv/bin/python scripts/sim_micro_parity.py  # cross-engine parity
```

3 Extended results figures

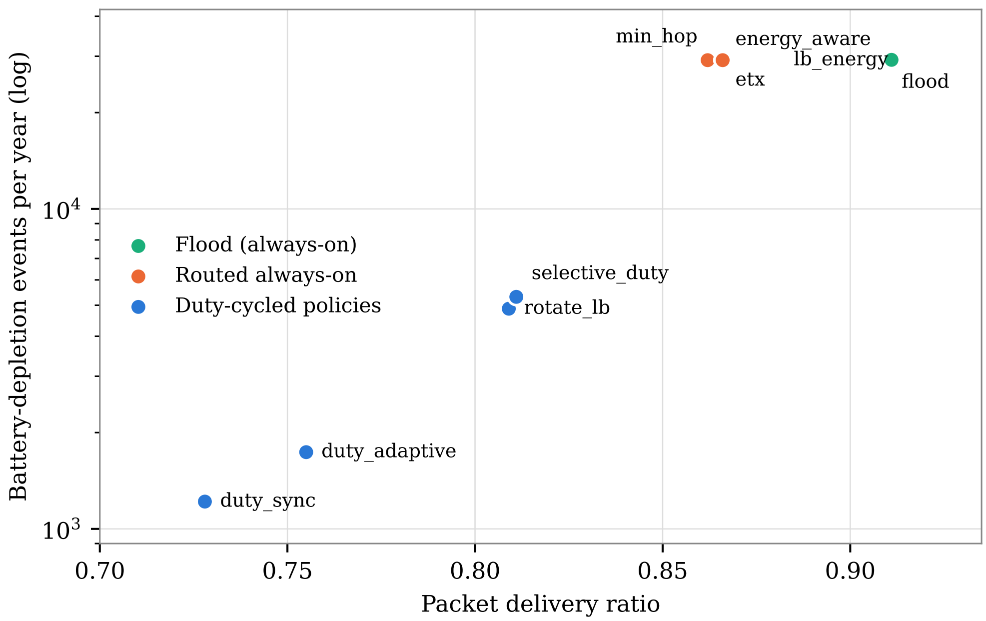

Figure 1: The survivability trade (MODEL-ONLY, uncalibrated; release_v1, five-seed means, PDR CIs < ±0.003). The five always-on routing modes cluster indistinguishably at 29,164–29,234 depletions per year across a 0.05 span of delivery ratio; duty-cycled policies occupy a separate regime, 5.5–24× fewer depletions at 0.05–0.14 lower delivery ratio. No overall winner is declared; the frontier is the finding.

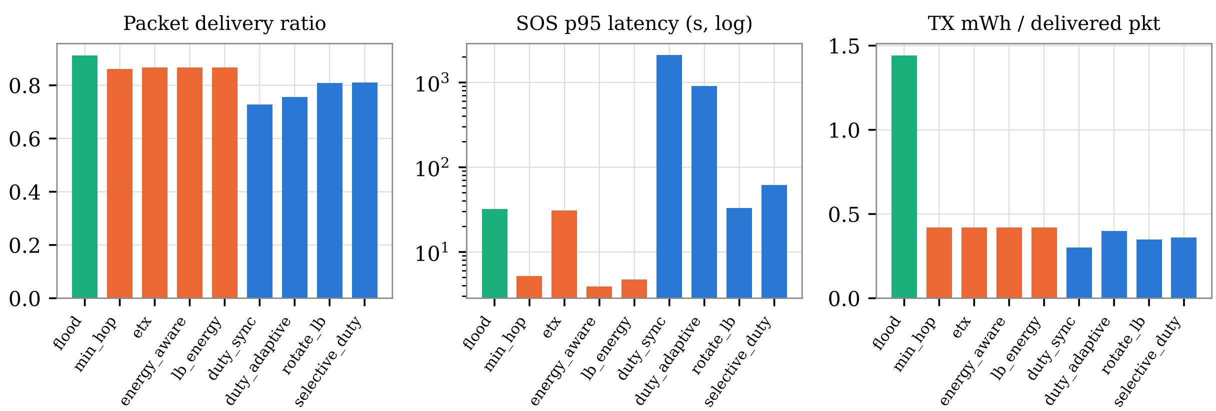

Figure 2: The cost side of duty-cycling (MODEL-ONLY): delivery ratio, SOS 95th-percentile latency (log scale; duty_sync's 94.7% delivery conceals a 35-minute p95 while backbone-awake duty modes hold 33–62 s), and transmit energy per delivered packet.

4 Full results table

Mode	PDR	±CI	Deaths/yr	±CI	SOS del.	SOS p95	Avail.	mWh/pkt
flood	0.911	0.0001	29,234	1.4	100%	32 s	53.5%	1.44
min_hop	0.862	0.0002	29,164	1.4	100%	5 s	53.7%	0.42
etx	0.866	0.0002	29,165	2.0	100%	31 s	53.7%	0.42
energy_aware	0.866	0.0002	29,164	2.1	100%	4 s	53.7%	0.42
lb_energy	0.866	0.0002	29,164	1.4	99.9%	5 s	53.7%	0.42
duty_sync	0.728	0.0002	1,217	1.1	94.7%	35.0 min	96.5%	0.30
duty_adaptive	0.755	0.0028	1,737	0.9	99.3%	15.0 min	94.7%	0.40
rotate_lb	0.809	0.0002	4,877	2.3	99.5%	33 s	91.0%	0.35
selective_duty	0.811	0.0001	5,311	5.3	99.3%	62 s	89.7%	0.36

Table 2: Complete release_v1 results (MODEL-ONLY, uncalibrated): nine modes × seeds 42–46 × 365 days; means with t-based 95% confidence half-widths. Source: `corrected_stats.json`, `meaningful_metrics.json`.